

# CAD-first 3-D library + bat rundown (v0.90.0)

The CAD image is now the face of every part. The interactive 3-D model and the cited manual figure are one click away.

## WHAT YOU SEE NOW

GRID every card

- leads with the CAD image (/cadimg)
- textured + coloured + dimensioned
- detail follows the build tier (&tier=)
- parametric SVG = loading placeholder only

uniform CAD face across the library

→

MODAL click a part → tabs (CAD opens first)

-  CAD image (default) — the rendered representative CAD, scaled to FLIS dims
-  Interactive 3-D — the live WebGL parametric model (spin / zoom)
-  Manual illustration — the real cited TM scan, when one exists
-  Approximation — opt-in experimental image→3-D

CAD at the forefront; figure + live 3-D secondary

## RUN THESE — every CAD/3-D bat (all compile-verified)

| BAT                | WHAT IT DOES                                                                                | WHEN                                          |
|--------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------|
| RUN-CAD-TIERS.bat  | Render ALL THREE tiers (v1/v2/v3) for the whole ~32,622-part set — the complete collection. | ★ run this for a complete CAD set (resumable) |
| MAKE-CAD.bat       | Render only the modern v3 (textured) set into index/cadcache.                               | modern set only                               |
| RUN-CAD-BATCH.bat  | Stop any running batch, clear orphaned older renders, render v3 fresh.                      | clean restart of v3                           |
| CAD-STATUS.bat     | Live progress: per-tier counts (legacy/lite/modern), %, rate, ETA, or COMPLETE.             | check progress anytime                        |
| RENDER-CONTACT.bat | A 10-part textured contact sheet -> docs/cad_contact_sheet.png.                             | see the textures                              |
| RENDER-COMPARE.bat | A 50-part v1-vs-v2 comparison sheet -> docs/cad_v1_vs_v2.png.                               | see the v1→v2 jump                            |
| RUN-CADCHECK.bat   | Start a throwaway server on 8766 and HTTP-probe /cadimg -> a real PNG.                      | smoke-test the route                          |

Don't have a cached image? The server renders it on demand: GET /cadimg?nsn=... (or ?style=v1|v2|v3 / ?tier=modern|lite|legacy). STL/OBJ at /cadstl /cadobj. Tip: never click inside a running batch console — that pauses it (press Esc to resume).